27/03/2025

CODE-A

Aakash

Medical | IIT-JEE | Foundations

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

AIM – 720

(Advanced INTENSIVE Mastery for 720)

MM : 720

PST-1

Time : 180 Mins.

Answers

1. (3)	37. (3)	73. (3)	109. (1)	145. (1)
2. (1)	38. (3)	74. (3)	110. (4)	146. (4)
3. (3)	39. (4)	75. (3)	111. (2)	147. (2)
4. (2)	40. (2)	76. (2)	112. (2)	148. (4)
5. (2)	41. (4)	77. (2)	113. (2)	149. (1)
6. (3)	42. (1)	78. (3)	114. (2)	150. (1)
7. (3)	43. (3)	79. (1)	115. (2)	151. (2)
8. (2)	44. (1)	80. (2)	116. (3)	152. (2)
9. (2)	45. (3)	81. (1)	117. (1)	153. (2)
10. (3)	46. (1)	82. (2)	118. (3)	154. (4)
11. (4)	47. (4)	83. (2)	119. (3)	155. (4)
12. (3)	48. (2)	84. (2)	120. (4)	156. (4)
13. (2)	49. (3)	85. (4)	121. (2)	157. (3)
14. (2)	50. (1)	86. (4)	122. (2)	158. (3)
15. (1)	51. (3)	87. (1)	123. (3)	159. (1)
16. (2)	52. (2)	88. (4)	124. (2)	160. (4)
17. (4)	53. (1)	89. (3)	125. (4)	161. (2)
18. (1)	54. (2)	90. (3)	126. (3)	162. (4)
19. (1)	55. (2)	91. (3)	127. (4)	163. (2)
20. (3)	56. (2)	92. (3)	128. (3)	164. (3)
21. (3)	57. (2)	93. (3)	129. (1)	165. (3)
22. (4)	58. (4)	94. (2)	130. (2)	166. (2)
23. (2)	59. (4)	95. (1)	131. (3)	167. (4)
24. (1)	60. (3)	96. (1)	132. (3)	168. (1)
25. (3)	61. (2)	97. (2)	133. (2)	169. (2)
26. (1)	62. (2)	98. (2)	134. (2)	170. (1)
27. (1)	63. (4)	99. (3)	135. (3)	171. (3)
28. (1)	64. (2)	100. (2)	136. (4)	172. (4)
29. (2)	65. (1)	101. (3)	137. (1)	173. (1)
30. (1)	66. (2)	102. (1)	138. (2)	174. (3)
31. (1)	67. (2)	103. (4)	139. (3)	175. (4)
32. (2)	68. (1)	104. (1)	140. (2)	176. (3)
33. (2)	69. (3)	105. (3)	141. (3)	177. (1)
34. (2)	70. (2)	106. (3)	142. (4)	178. (2)
35. (3)	71. (1)	107. (3)	143. (1)	179. (1)
36. (2)	72. (2)	108. (2)	144. (3)	180. (4)

Hints and Solutions

PHYSICS

(1) **Answer :** (3)
Solution:
Path difference = $\frac{\lambda}{2\pi}$× Phase difference.
For fifth minimum, phase difference = 9π
$\therefore \quad \Delta x = \frac{\lambda}{2\pi} \times 9\pi = \frac{9\lambda}{2}$

(2) **Answer :** (1)
Solution:
For first minimum $a\sin\theta = \lambda$
Path difference between mid-point wavelet and edge wavelet
= $\frac{1}{2}(a\sin\theta) = \frac{\lambda}{2}$
Phase difference = $\frac{2\pi}{\lambda} \times \frac{\lambda}{2}$
= π radian

(3) **Answer :** (3)
Solution:
Photoelectric current is directly proportional to intensity of incident light. With increases in intensity of incident light, photo electric current increases.

(4) **Answer :** (2)
Solution:
de Broglie wavelength $\lambda = \frac{h}{\sqrt{2mk}}$
$\lambda' = \frac{h}{\sqrt{2m \cdot 16k}} = \frac{\lambda}{4}$
$|\%\,\Delta\lambda| = \left|\frac{\lambda - \lambda'}{\lambda}\right| \times 100 = 75\%$

(5) **Answer :** (2)
Solution:
Distance of closest approach
$r_0 = \frac{Ze^2}{2\pi\varepsilon_0 mv^2} \quad \therefore\ r_0 \propto \frac{1}{m}$

(6) **Answer :** (3)
Solution:
$Y = \overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = A + B$

(7) **Answer :** (3)
Solution:
$E = \frac{hc}{\lambda} = \frac{6.63\times10^{-34}\times3\times10^{8}}{975\times10^{-10}\times1.6\times10^{-19}} = 12.75$ eV
After absorbing 12.75 eV, electron jump to 3rd excited state (n = 4).
No. of spectral lines emitted = $n\frac{(n-1)}{2}$
= $\frac{4(4-1)}{2} = 6$

(8) **Answer :** (2)
Solution:
$\lambda_{electron} = \lambda_{proton}$
$E_{photon} = \frac{hc}{\lambda_{photon}}$ and $p_{electron} = \frac{h}{\lambda_{electron}}$
$\frac{E_{photon}}{p_{electron}} = \frac{hc}{\lambda_{photon}} \times \frac{\lambda_{electron}}{h} = c$

(9) **Answer :** (2)

Solution:

Distance of a bright fringe from central maxima is $y_n = \frac{nD\lambda}{d}$

∴ $n\lambda$ = constant at a given point

$n_1\lambda_1 = n_2\lambda_2$

$\frac{16\times7200}{4800} = n_2 = 24$

(10) **Answer :** (3)

Solution:

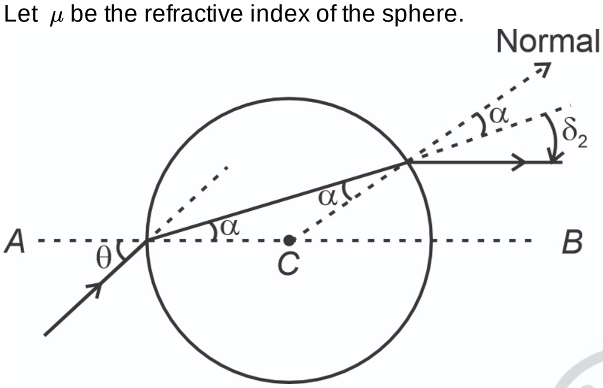
${}^{A}_{Z}X \xrightarrow{\alpha} {}^{A-4}_{Z-2}Y \xrightarrow{2\beta^-} {}^{A-4}_{Z}Z$

∴ X and Z are isotopes.

(11) **Answer :** (4)

Solution:

Let μ be the refractive index of the sphere.



$1 \times \sin\theta = \mu \times \sin\alpha$ (i)

For emergent ray to be parallel, $\delta_1 + \delta_2 = \theta$

$\delta_1 = \theta - \alpha$, $\delta_2 = \theta - \alpha$

$2(\theta - \alpha) = \theta \Rightarrow 2\theta - 2\alpha = \theta \quad \alpha = \frac{\theta}{2}$

Putting this in equation (i), we get

$1 \times \sin\theta = \mu\sin\frac{\theta}{2}$

$2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = \mu\sin\frac{\theta}{2}$

$\Rightarrow \quad \mu = 2\cos\frac{\theta}{2}$

(12) **Answer :** (3)

Solution:

On comparing with $B = B_0\sin(kx - \omega t)$

$K = \frac{2\pi}{\lambda} = 5000\pi$

$\lambda = \frac{2}{5000}$

$= 0.4 \times 10^{-3} = 0.4$ mm.

(13) **Answer :** (2)

Solution:

$u = -10$ cm $v = ?$ $f = +12$ cm

Using $-\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

$\frac{1}{10} + \frac{1}{v} = \frac{1}{12}$

$\frac{1}{v} = \frac{1}{12} - \frac{1}{10}$

$v = -60$ cm, virtual

(14) **Answer :** (2)

Solution:

Kinetic energy = – (Total energy).

(15) **Answer :** (1)

Solution:

Fringe width $\beta = \frac{D\lambda}{d}$

$\frac{\beta_1}{\beta_2} = \frac{\lambda_1}{\lambda_2} \frac{D_1}{D_2} \frac{d_2}{d_1}$

$\Rightarrow \ \frac{D_1}{D_2} = \frac{\beta_1}{\beta_2} \frac{\lambda_1}{\lambda_2} \frac{d_1}{d_2}$

$\Rightarrow \ \frac{D_1}{D_2} = \left(\frac{1}{1}\right)\left(\frac{2}{1}\right)\left(\frac{2}{1}\right) = \frac{4}{1}$

(16) Answer : (2)

Solution:

As per law of conservation of linear momentum

$m_1 v_1 = m_2 v_2$

$\frac{m_1}{m_2} = \frac{v_2}{v_1} = \frac{1}{2}$

As nuclear density is constant $\therefore m \propto r^3$.

$\frac{m_1}{m_2} = \frac{1}{2} = \frac{r_1^3}{r_2^3}$

$\therefore \quad \frac{r_1}{r_2} = \frac{1}{2^{1/3}}$

(17) Answer : (4)

Solution:

Total energy of EM wave in vacuum is $\frac{1}{4}\varepsilon_0 E^2 + \frac{1}{4}\frac{B^2}{\mu_0}$

(18) Answer : (1)

Solution:

1. For real and inverted image $m = -2$

$m = \frac{f}{f - u}$

$-2 = \frac{-50}{-50 - u}$

$\therefore u = -75$ cm

(19) Answer : (1)

Solution:

$m = 1 + \frac{D}{f}$

$= 1 + \frac{25}{5} = 6$

(20) Answer : (3)

Solution:

L.C = $\frac{\text{Pitch}}{\text{No of divisions on circular scale}}$

Pitch = L.C × Circular scale divisions

$= 0.5 \times 10^{-3} \times 100 = 5 \times 10^{-2}$ m

= 5 cm

(21) Answer : (3)

Solution:

Let I be incident unpolarised light. Then intensity of the polarised light through polariser is $\frac{I}{2}$. Using law of Malus, intensity of transmitted light through analyser will be

$4I_0 = \frac{I}{2}\cos^2 45^\circ$

$4I_0 = \frac{I}{2}\left(\frac{1}{\sqrt{2}}\right)^2$

$I = 16I_0$

(22) Answer : (4)

Solution:

Focal length of the mirror is independent of medium.

(23) Answer : (2)

Solution:

$P = P_1 + P_2$

$\frac{100}{60} = \frac{100}{25} + P_2$

1.67 D – 4 D = –2.33 D

(24) **Answer :** (1)
Solution:
Nuclear force is charge independent. Stability of nucleus is determined by binding energy per nucleon.

(25) **Answer :** (3)
Solution:
Lyman series constitute spectral lines corresponding to transition from higher energy to ground state of hydrogen atom.

(26) **Answer :** (1)
Solution:
Work function $\phi_0 = h\nu$

$2h\nu = \phi_0 + \frac{1}{2}mv^2_{\text{max}}$

$v^2_{\text{max}} = \frac{2(2h\nu - h\nu)}{m} = \frac{2h\nu}{m}$

$v_{\text{max}} = \sqrt{\frac{2h\nu}{m}}$

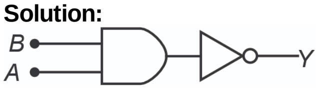
(27) **Answer :** (1)
Solution:
Second excited state means $n = 3$.
Therefore, energy needed to ionize = $\frac{13.6}{3^2}$eV
= 1.51 eV

(28) **Answer :** (1)
Solution:
Radiation pressure on a purely reflecting surface, $P = \frac{2I}{c}$
Force = pressure × area
For equilibrium :
spring force = force due to radiation

$2Kx_0 = \frac{2IA}{c}$

$x_0 = \frac{IA}{Kc}$

(29) **Answer :** (2)
Solution:
Output frequency remains same in half wave rectifier.

(30) **Answer :** (1)
Solution:


A	B	$\overline{A \cdot B}$	Y
0	0	$\bar{0}$	1
1	0	$\bar{0}$	1
0	1	$\bar{0}$	1
1	1	$\bar{1}$	0

(31) **Answer :** (1)
Solution:
(n + 1) VSD = n MSD
VSD = $\frac{n}{n+1}$MSD
L.C = MSD – VSD
= MSD $\left[\frac{n+1-n}{n+1}\right] = \frac{x}{n+1}$

(32) **Answer :** (2)
Solution:

First secondary maximum of the diffraction pattern:

$d\sin\theta = \frac{3}{2}\lambda$

$d = \frac{3}{2} \times \frac{6500\times10^{-10}}{\sin 30^\circ}$

$= \frac{3}{2} \times \frac{2}{1} \times 6500 \times 10^{-10}$

= 1.95 × 10^{-6}

= 1.95 µm.

(33) Answer : (2)

Solution:

$\nu = \frac{v}{2L} = 2800$ (for open pipe)

$\nu' = \frac{v}{4L'}$ (for closed pipe)

L' = 0.8 L

$\frac{v}{2L} = 2800$ ...(i)

$\frac{v}{4(0.8L)} = \nu'$ ...(ii)

1.6 = $\frac{2800}{\nu'}$

$\nu' = \frac{2800}{1.6}$

= 1750 Hz

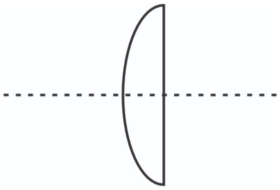
(34) Answer : (2)

Solution:

Momentum of EM wave , $P = \frac{E}{C}$

(35) Answer : (3)

Solution:



$\frac{1}{f} = \left(\mu - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

R_1 = 18 cm, R_2 = ∞

$\frac{1}{f} = (1.6 - 1)\left(\frac{1}{18}\right)$

$\frac{1}{f} = \frac{0.6}{18}$, f = 30 cm

(36) Answer : (2)

Solution:

B.E. per nucleon of ${}^{7}_{3}$Li

= $\frac{39.3}{7}$ MeV

= 5.61 MeV

B.E. per nucleon of ${}^{4}_{2}$He

= $\frac{27.37}{4}$ MeV

= 6.84 Mev

B.E. per nucleon of ${}^{56}_{26}$Fe

= $\frac{492.26}{56}$ = 8.790 MeV

As B.E. per nucleon is minimum for ${}^{7}_{3}$Li. Hence it is least stable.

(37) Answer : (3)

Solution:

For balanced bridge

$\frac{R_1}{R_2} = \frac{L_1}{L_2}$, $\frac{R_1}{R_2} = \frac{20}{80} = \frac{1}{4}$

(38) Answer : (3)

Solution:

$\lambda = \frac{h}{\sqrt{3mKT}}$

$\lambda \propto \frac{1}{\sqrt{T}}$

$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{T_2}{T_1}}$

$\Rightarrow \quad \frac{\lambda}{\lambda'} = \sqrt{\frac{600}{300}} = \sqrt{2}$

$\lambda' = \frac{\lambda}{\sqrt{2}}$

(39) Answer : (4)
Solution:

$\lambda = \frac{h}{\sqrt{2mE}}$

$\lambda \propto \frac{1}{\sqrt{E}}$

$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{E_2}{E_1}} \Rightarrow \quad \frac{10^{-10}}{0.25 \times 10^{-10}} = \frac{4}{1}$

$\frac{E_2}{E_1} = \frac{16}{1} \Rightarrow E_2 = 16E_1$

Added energy = $16E_1 - E_1 = 15E_1$

(40) Answer : (2)
Solution:
Matter waves are not EM waves

(41) Answer : (4)
Solution:
• Convex mirror have large field view, therefore used as side view mirror. It also produce erect and diminished image.
• Concave mirrors are used by dentists.
• When real object is place near the plane mirror, it forms virtual, erect and same size image.

(42) Answer : (1)
Solution:
In hydrogen atom

$r = \frac{r_0 \times n^2}{Z} \Rightarrow r \propto n^2$

$v = \frac{v_0 \times Z}{n} \Rightarrow v \propto \frac{1}{n}$

$T = \frac{2\pi r}{v} \Rightarrow T \propto \frac{\frac{n^2}{z}}{\frac{z}{n}} \Rightarrow T \propto n^3$

$B = \frac{\mu_0 \mathrm{i}}{2r} = \frac{\mu_0 e}{2rT} \quad \Rightarrow B \propto \frac{1}{n^2 \times n^3} \quad \Rightarrow B \propto \frac{1}{n^5}$

(43) Answer : (3)
Solution:
In the given emission, momentum of the system remain conserved.

$\underset{\substack{\text{Mother} \\ \text{nucleus}}}{X} \rightarrow \underset{\substack{\text{Daughter} \\ \text{nucleus}}}{\mathrm{Y}} + \alpha - \text{particle}$

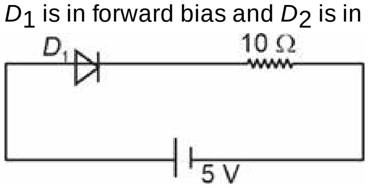
$(KE)_Y$ = 2 MeV

$(KE)_\propto$ = 50 MeV – 2 MeV = 48 MeV

$KE = \frac{P^2}{2m} \quad \Rightarrow \quad \frac{2}{48} = \frac{M_\propto}{M_y}$

$M_y = 96$

(44) Answer : (1)
Solution:
D_1 is in forward bias and D_2 is in reverse bias.



$I = \frac{5}{10} = 0.5$ A

(45) Answer : (3)
Solution:
Shortest wavelength of Balmer series of hydrogen-like atom

$\frac{1}{\lambda_1} = RZ^2\left(\frac{1}{2^2} - \frac{1}{\infty}\right) \Rightarrow \frac{1}{\lambda_1} = \frac{RZ^2}{4}$ ...(i)

Shortest wavelength of Lyman series of hydrogen atom

$\frac{1}{\lambda_2} = R\left(\frac{1}{1} - \frac{1}{\infty}\right) \Rightarrow \frac{1}{\lambda_2} = R$ ...(ii)

Given $\lambda_1 = \lambda_2$

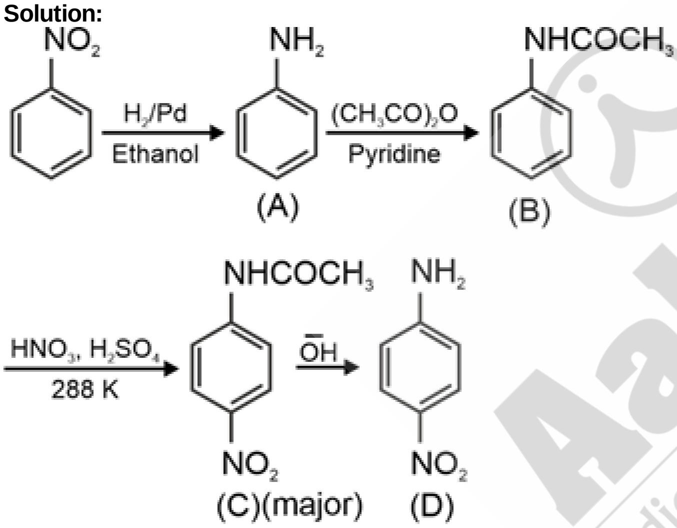
$\therefore \frac{RZ^2}{4} = R \Rightarrow Z = 2$

CHEMISTRY

(46) Answer : (1)
Solution:
Complete hydrolysis of DNA yields a pentose sugar, phosphoric acid and nitrogen containing heterocyclic compound called bases.

(47) Answer : (4)
Solution:
Serine and Tyrosine are non-essential amino acids.

(48) Answer : (2)
Solution:



(49) Answer : (3)
Solution:
Aliphatic and aromatic primary amines on heating with chloroform and ethanolic KOH form carbylamines.

$\mathrm{R{-}NH_2} \xrightarrow[\mathrm{KOH}\ \Delta]{\mathrm{CHCl_3}} \mathrm{R{-}NC}$

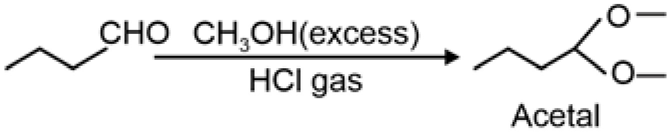
(50) Answer : (1)
Solution:

Compounds	**pK_b**
Benzenamine	9.38
N, N-Dimethylaniline	8.92
Phenylmethanamine	4.70

(51) Answer : (3)
Solution:
Boiling point of propan-1-ol is higher than acetone because intermolecular hydrogen bond is present in propan-1-ol as a result association of the molecules increases.

(52) Answer : (2)

Solution:



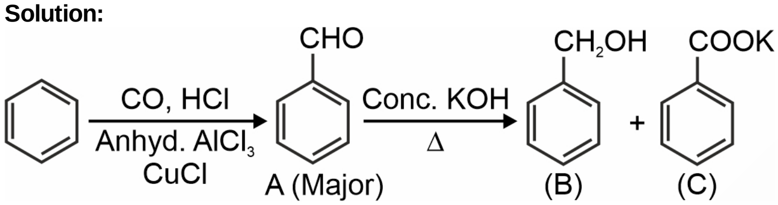
(53) **Answer :** (1)
Solution:

$R-\overset{\overset{\large O}{\|}}{C}-NH_2 + Br_2 + 4NaOH$

$\rightarrow R-NH_2 + Na_2CO_3 + 2NaBr + 2H_2O$

Hoffmann bromamide degradation reaction.

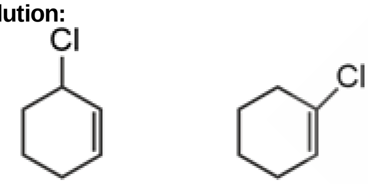
(54) **Answer :** (2)
Solution:



(55) **Answer :** (2)
Solution:
Fehling solution A is aqueous copper sulphate and Fehling solution B is alkaline sodium potassium tartarate.

(56) **Answer :** (2)
Solution:
A mixture containing two enantiomers in equal proportions will have zero optical rotation, as the rotation due to one isomer will be cancelled by the rotation due to other isomer.

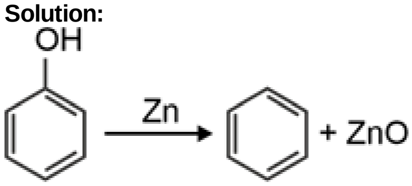
(57) **Answer :** (2)
Solution:



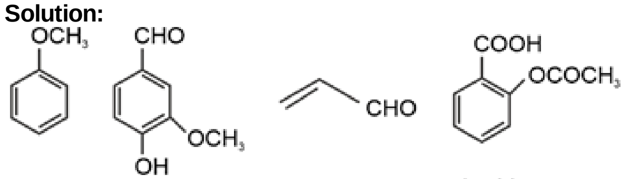
Allylic chloride Vinylic chloride

(58) **Answer :** (4)
Solution:
• Substitution reaction in allyl chloride is accelerated by π conjugation in transition state. Benzyl chloride a bit more reactive than allyl: benzene ring slightly better at π conjugation than isolated double bond.
• Substitution reaction in secondary alkyl chloride is slow because of steric hindrance.

(59) **Answer :** (4)
Solution:



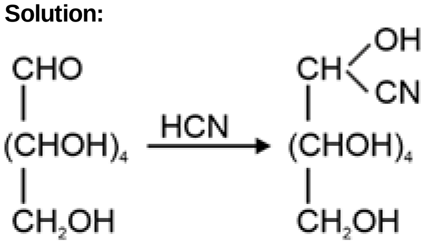
(60) **Answer :** (3)
Solution:



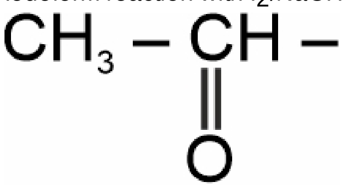
Anisole Vanillin Acrolein Aspirin

(61) **Answer :** (2)

Solution:

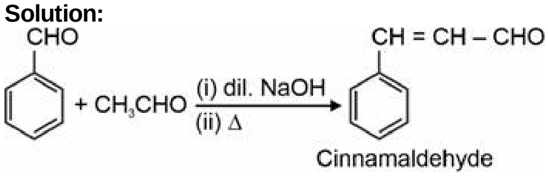


(62) **Answer :** (2)
Solution:
Iodoform reaction with I_2/NaOH is used for detection of CH_3CHO group or



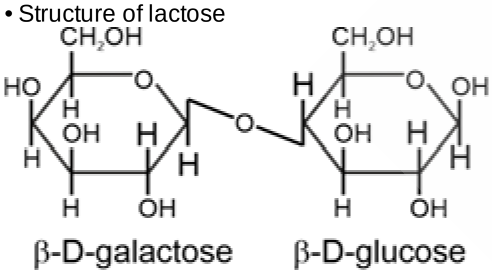
group which produces CH_3CO group on oxidation

(63) **Answer :** (4)
Solution:



(64) **Answer :** (2)
Solution:
Benzenediazonium chloride is a colourless crystalline solid and is readily soluble in water.

(65) **Answer :** (1)
Solution:
• Structure of lactose



(66) **Answer :** (2)
Solution:

Vitamin		Deficiency disease
Vitamin E	→	Increased fragility of RBCs
Pyridoxine	→	Convulsions
Riboflavin	→	Digestive disorders
Vitamin K	→	Increased blood clotting time

(67) **Answer :** (2)
Solution:

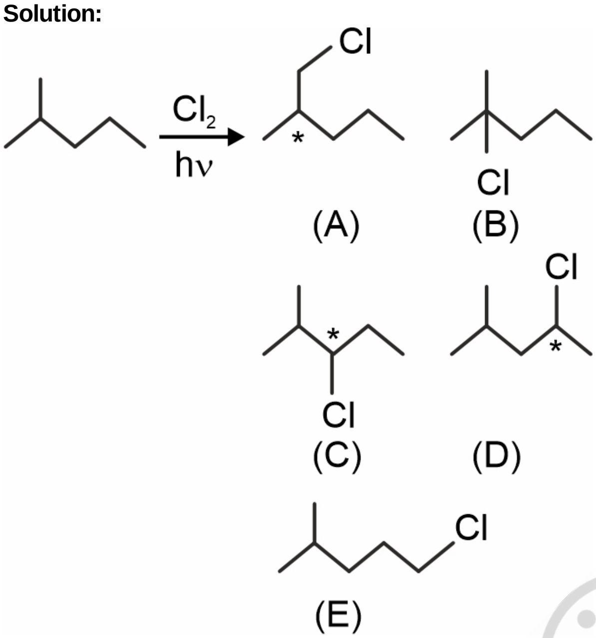
$$CH_3-C\equiv N \xrightarrow[\text{(ii) } H_3O^+]{\text{(i) PhMgBr}} CH_3-\overset{\overset{\displaystyle O}{||}}{C}-Ph \xrightarrow{H_2CrO_4} Ph-COOH$$

(68) **Answer :** (1)
Solution:
Correct order of acidic strength:

$BrCH_2COOH > HCOOH > ClCH_2CH_2COOH > C_6H_5COOH$

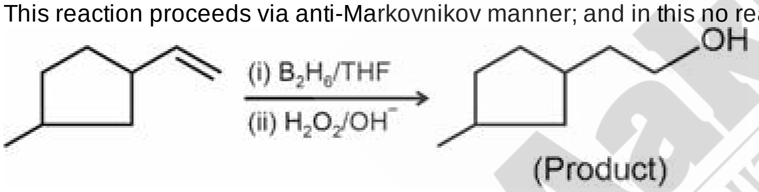
(69) Answer : (3)
Solution:
Tertiary alcohol on reaction with copper at 573 K gives alkene as major product.

(70) Answer : (2)
Solution:

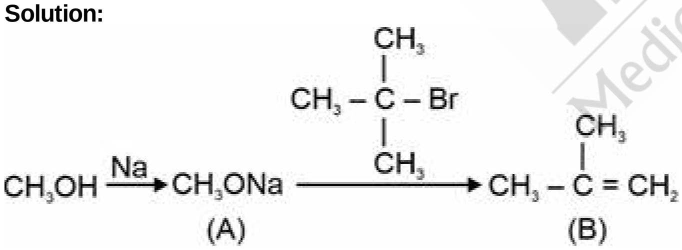


• (A), (B) and (D) will exist as a pair of enantiomers.
• Total number of compounds (including stereoisomers) obtained = 8

(71) Answer : (1)
Solution:
This reaction proceeds via anti-Markovnikov manner; and in this no rearrangement is possible.



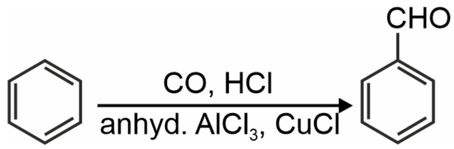
(72) Answer : (2)
Solution:



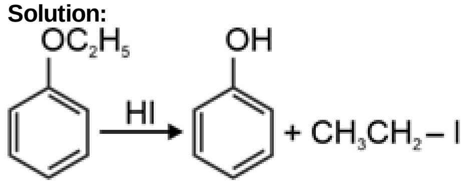
(73) Answer : (3)
Solution:
Thyroxine produced in thyroid gland is an iodinated derivative of amino acid tyrosine.

(74) Answer : (3)
Solution:
Electron withdrawing group present at para position increases the electrophilicity of carbonyl carbon and rate of ester hydrolysis increases.

(75) Answer : (3)
Solution:
Wurtz reaction:
$2RX + 2Na \rightarrow R{-}R + 2NaX$
• Gatterman-Koch reaction:



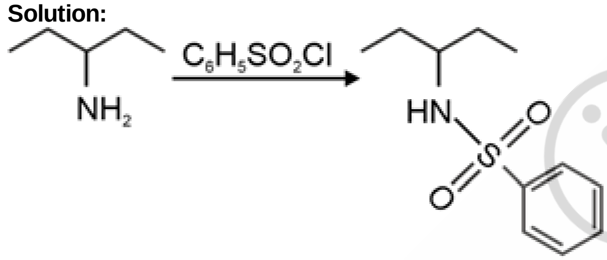
(76) Answer : (2)
Solution:



(77) Answer : (2)
Solution:

$$HCHO \xrightarrow[\text{(ii) } H_2O]{\text{(i) } C_2H_5MgBr} \underset{(A)}{CH_3CH_2CH_2OH} \xrightarrow{CrO_3 - H_2SO_4} \underset{(B)}{CH_3CH_2COOH}$$

(78) Answer : (3)
Solution:



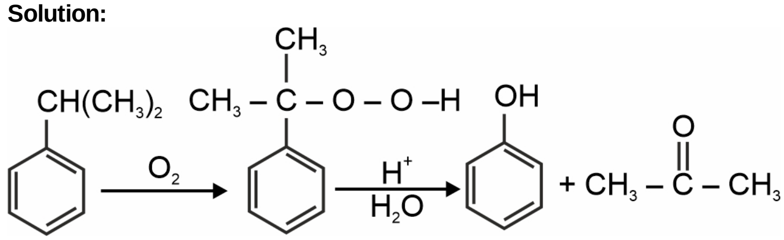
(Soluble in aqueous alkali)

(79) Answer : (1)
Solution:
Amylose is long unbranched chain with 200-1000 α-D– (+)-glucose units held together by $C_1 - C_4$ glycosidic linkage.

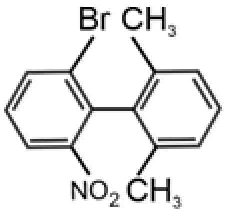
(80) Answer : (2)
Solution:

Compound	**Boiling point (K)**
n–$C_4H_9NH_2$	350.8
$(C_2H_5)_2NH$	329.3
$C_2H_5N(CH_3)_2$	310.5

(81) Answer : (1)
Solution:

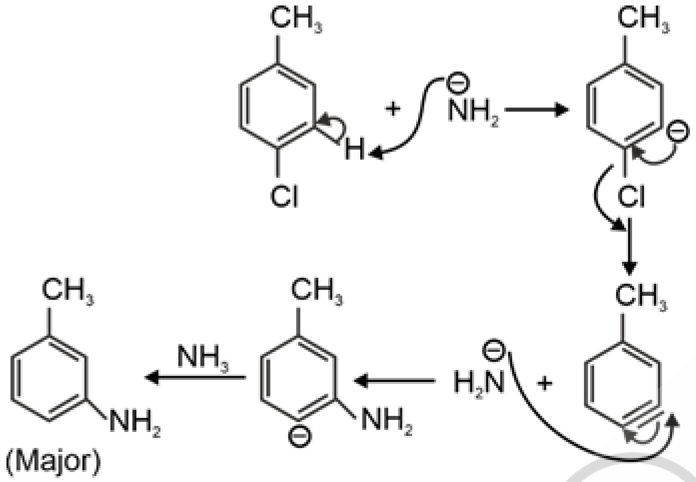


(82) Answer : (2)
Solution:



contains plane of symmetry, hence it is optically inactive compound.

(83) **Answer :** (2)
Solution:

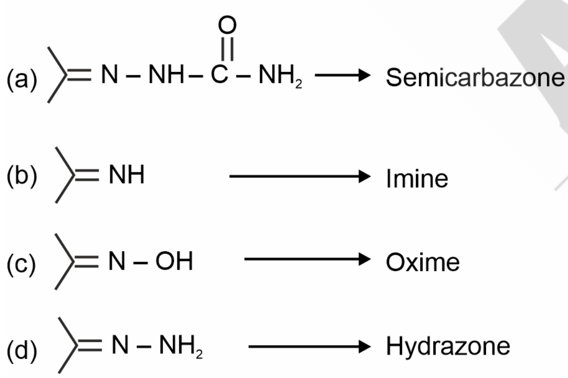


(84) **Answer :** (2)
Solution:
• Correct order of nucleophilicity in ethanol is:

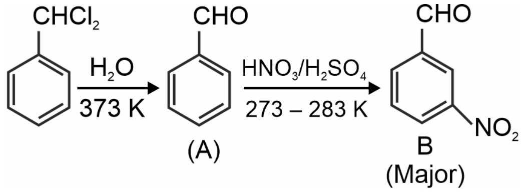
$I^- > EtO^- > PhO^-$

• EtO^- is a better nucleophile than PhO^- because in PhO^- the negative charge is delocalised in the benzene ring as a result its nucleophilicity decreases.

(85) **Answer :** (4)
Solution:
Carbonyl derivative **Name**



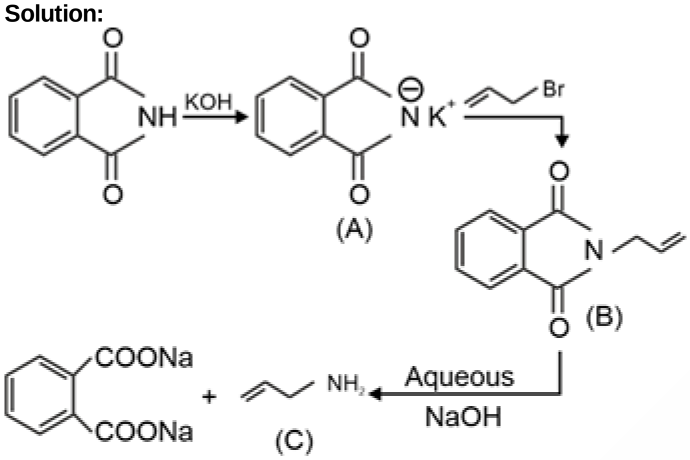
(86) **Answer :** (4)
Solution:



(87) **Answer :** (1)
Solution:
Acids stronger than carbonic acid liberate CO_2 gas on reaction with $NaHCO_3$.

(88) **Answer :** (4)
Solution:
Aromatic carboxylic acid does not undergo Friedel-Crafts reaction because the carboxyl group is deactivating and the catalyst aluminium chloride gets bonded to the carboxyl group.

(89) **Answer :** (3)
Solution:
• Primary alcohols give red colour in Victor Meyer's test.
• Secondary alcohol give blue colour in Victor Meyer's test.

(90) **Answer :** (3)
Solution:


BOTANY

(91) **Answer :** (3)
Solution:
Cuscuta is a non-green plant which lives as a parasite on hedge plants.

(92) **Answer :** (3)
Solution:
In order to avoid interspecific competition, many species have evolved the mechanism of resource partitioning by choosing the different foraging patter.

(93) **Answer :** (3)
Solution:
Birth rate = $\frac{8500-850}{850} = 9$

(94) **Answer :** (2)
Solution:
The ladybird beetle is useful to get rid of aphids.

(95) **Answer :** (1)
Solution:
Penicillin was discovered by Alexander Fleming while working on *Staphylococci* bacteria.

(96) **Answer :** (1)
Solution:
During growth, the LAB produce acids that coagulate and partially digest the milk proteins.

(97) **Answer :** (2)
Solution:
Large holes in ‘Swiss cheese’ are due to the production of a large amount of CO_2 by a bacterium *Propionibacterium sharmanii.*

(98) **Answer :** (2)
Solution:
Statins are the blood-cholesterol lowering agents which are produced by the yeast, *Monascus purpureus.*

(99) **Answer :** (3)

Solution:
Forest is a terrestrial ecosystem while rest of them are aquatic ecosystems.

(100) Answer : (2)
Solution:
Anthropogenic ecosystem is a man-made ecosystem. It does not possess a self-regulatory mechanism.

(101) Answer : (3)
Solution:

1. By the process of leaching, water soluble inorganic nutrients go down into the soil horizon and get precipitated as unavailable salts.

(102) Answer : (1)
Solution:
GFC is the major conduit of energy flow in an aquatic ecosystem.

(103) Answer : (4)
Solution:
In an ecosystem, there is unidirectional movement of energy towards the higher trophic levels from the lower trophic level.

(104) Answer : (1)
Solution:
Mortality refers to the number of individuals that have died during a given period in a population.

(105) Answer : (3)
Solution:
The bases of the two strands of DNA are paired through hydrogen bonds.

(106) Answer : (3)
Solution:
Given figure is of a nucleosome where **A** is histone octamer and **B** is DNA.

(107) Answer : (3)
Solution:
Erwin Chargaff observed that the ratios between Adenine and Thymine and Guanine and Cytosine are constant and equals one.

(108) Answer : (2)
Solution:
Uracil is present in RNA at the place of Thymine.

(109) Answer : (1)
Solution:
Observations from Griffith's experiment are as follows:
- S strain bacteria (heat-killed) → Injected into mice → Mice lived
- S strain bacteria (heat-killed) + R strain (live) → Injected into mice → Mice died
- S strain (live) → Injected into mice → Mice died
- R strain (live) → Injected into mice → Mice lived

(110) Answer : (4)
Solution:
RNA, being unstable, mutates at a faster rate than DNA.

(111) Answer : (2)
Solution:
The template strand has the polarity 3'→5' and is also known as the non-coding strand. The presence of the promoter sequence is towards the 3' end of the template strand.

(112) Answer : (2)
Solution:
In Meselson and Stahl experiment, the DNA that was extracted from the 1st generation of *E. coli* (after 20 minutes) had a hybrid or intermediate density due to the presence of both heavy and light DNA.

(113) Answer : (2)
Solution:
In eukaryotes, the regulation of gene expression may occur at the transcriptional level, processing level (regulation of splicing), transport of mRNA from nucleus to the cytoplasm and translational level.

(114) Answer : (2)

Solution:
In bacteria, there is no separation of cytosol and nucleus and hence, transcription and translation take place in the same compartment.

(115) Answer : (2)
Solution:
The chemical method developed by Har Gobind Khorana was instrumental in synthesising RNA molecules with defined combination of bases.

(116) Answer : (3)
Solution:
The correct sequence on the mRNA will be 5'AUCGAUCGAUCGAUC 3' for the given template

(117) Answer : (1)
Solution:
SNPs stands for Single Nucleotide Polymorphisms.

(118) Answer : (3)
Solution:
Some of the salient features of human genome are as follows:
Less than 2 percent of the genome codes for protein.
SNPs (snips) occur at about 1.4 million locations.
Chromosome 1 has most genes (2968) and Y has the fewest.

(119) Answer : (3)
Solution:
Endoparasites have a complex life cycle because of their extreme specialization.

(120) Answer : (4)
Solution:
In the *lac* operon, the structural gene 'z' codes for the enzyme beta-galactosidase which is primarily responsible for the hydrolysis of lactose.

(121) Answer : (2)
Solution:
Curve 'a' in the figure represents population growth curve when resources are not limiting and hence the population grows exponentially. This is represented by the equation $\frac{dN}{dt} = rN$.

(122) Answer : (2)
Solution:
Growing population shows triangular age pyramid and stable population shows bell-shaped age pyramid.

(123) Answer : (3)
Solution:
Sea anemone and clown fish - Commensalism
Mycorrhizae - Mutualism
Marine fish and copepods - Parasitism
Abingdon tortoise and goat - Competition

(124) Answer : (2)
Solution:
TMV and aphids are not the biocontrol agents. Baculoviruses are the excellent candidates for narrow spectrum insecticidal applications.

(125) Answer : (4)
Solution:
Tertiary treatment is a physico-chemical process in which one of the methods used to remove harmful chemicals is reverse osmosis.

(126) Answer : (3)
Solution:
Sun is not the energy source for deep sea hydro-thermal ecosystem. Less than 50% of the total incident solar radiation is photosynthetically active.

(127) Answer : (4)
Solution:
Stratification is not a functional aspect of an ecosystem.

(128) Answer : (3)
Solution:

According to Lindemann's 10% law, the energy available to:
Grasshopper – 10% of 60 J = 6 J
Frog – 10% of 6 J = 0.6 J
Snake – 10% of 0.6 J = 0.06 J

(129) Answer : (1)
Solution:
If the sequence of bases in one strand is known, then the sequence in other strand can be predicted because both the strands of DNA are complementary to each other.

(130) Answer : (2)
Solution:
Histones are a set of positively charged basic proteins. Histone is rich in the basic amino acids residues lysine and arginine. They both carry positive charge.

(131) Answer : (3)
Solution:
An operator gene receives the product of regulator gene and also allows the functioning of the operon when it is not covered by the biochemical produced by regulator gene.

(132) Answer : (3)
Solution:
The nucleosomes in chromatin are seen as 'beads-on-string' structure when viewed under electron microscope.

(133) Answer : (2)
Solution:
The term 'Nuclein' was given by Friedrich Meischer.

(134) Answer : (2)
Solution:
RNA polymerase II transcribes for the precursor of mRNA, the heterogenous nuclear RNA (hnRNA).

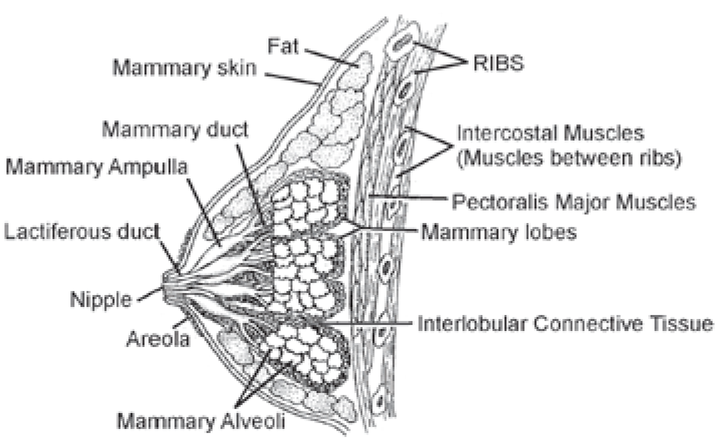
(135) Answer : (3)
Solution:
In predation, the predator gets benefit whereas the interaction is detrimental to the prey.

ZOOLOGY

(136) Answer : (4)
Solution:
At present, about 30 recombinant therapeutics have been approved for human use the world over. In India, 12 of these are presently being marketed.

(137) Answer : (1)
Solution:
Ovaries are the primary female sex organs that produce female gametes (ova) and several steroidal hormones (progesterone, estrogen, *etc.*). Uterus and vagina are the female accessory ducts.

(138) Answer : (2)
Solution:



(139) Answer : (3)
Solution:
Pomato is a somatic hybrid formed by the process of somatic hybridisation of potato and tomato while Golden rice is a transgenic crop. Rosie is the transgenic cow that produced human protein-enriched milk. Bt brinjal contains gene from *Bacillus thuringiensis* to produce protoxin.

(140) Answer : (2)
Solution:


(141) Answer : (3)
Solution:
Meloidogyne incognita is a nematode which infects the roots of tobacco plants and causes a great reduction in their yield. *E.coli, Haemophilus influenzae* and *Agrobacterium* are bacteria.

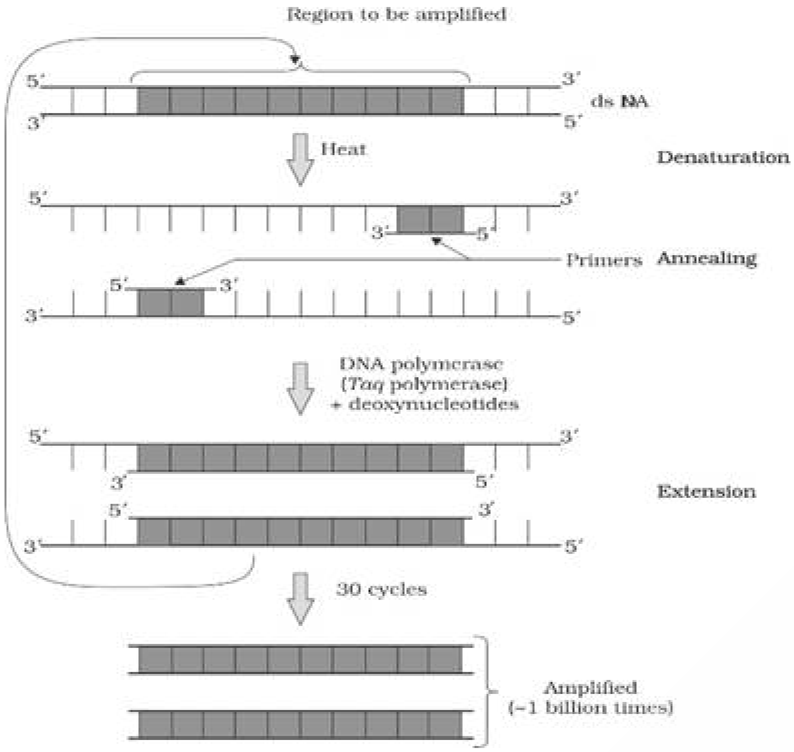
(142) Answer : (4)
Solution:
Secondary sexual characters develop in males due to androgens.
Males have low-pitched voice. Androgens are released by the Leydig cells.

(143) Answer : (1)
Solution:
For the construction of the first rDNA, native plasmid was taken from *Salmonella typhimurium*. This was accomplished in 1972 by Stanley Cohen and Herbert Boyer.

(144) Answer : (3)
Solution:
Nucleases are of two kinds; exonucleases and endonucleases. Exonucleases remove nucleotides from the ends of the DNA whereas, endonucleases make cut at specific positions within the DNA. Nucleases belong to class III of enzymes.

(145) Answer : (1)
Solution:
Micro-injection is a vector-less method of gene transfer. In this method, the recombinant DNA is directly injected into the nucleus of an animal cell. In biolistics or gene gun (suitable for plants), cells are bombarded with high velocity micro-particles of gold or tungsten coated with DNA.

(146) Answer : (4)
Solution:
Restriction endonucleases are obtained from prokaryotes.
Today, we know more than 900 restriction enzymes that have been isolated from over 230 strains of bacteria each of which recognises different recognition sequences.



(147) Answer : (2)
Solution:
Areola – Pigmented area surrounding the nipple on the mammary glands.
Hymen – Present in vagina
Fimbriae **–** Finger-like projections of oviducts

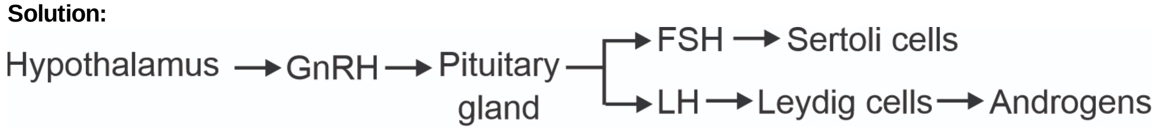
(148) Answer : (4)
Solution:
A primary spermatocyte completes the first meiotic division leading to the formation of two equal and haploid cells called the secondary spermatocytes. The secondary spermatocytes undergo the second meiotic division to produce four equal and haploid spermatids.
→ 5 primary spermatocytes × 2 = 10 secondary spermatocytes
→ 10 secondary spermatocytes × 2 ⇒ 20 spermatids

(149) Answer : (1)
Solution:
Chorionic villi are finger-like projections which appear on the trophoblast and get surrounded by the uterine tissue and maternal blood.
Fimbriae are finger-like projections present over the edges of infundibulum.

(150) Answer : (1)
Solution:
Lepidopterans – Tobacco budworms, army worms
Coleopterans – Beetles
Dipterans – Flies and mosquitoes

(151) Answer : (2)
Solution:
Isolated protoplasts from two different varieties of plants each having desirable characters can be fused to get the hybrid protoplast which can be further grown to form a new plant. These hybrids are called somatic hybrids, while the process is called somatic hybridisation.

(152) Answer : (2)
Solution:



(153) Answer : (2)
Solution:

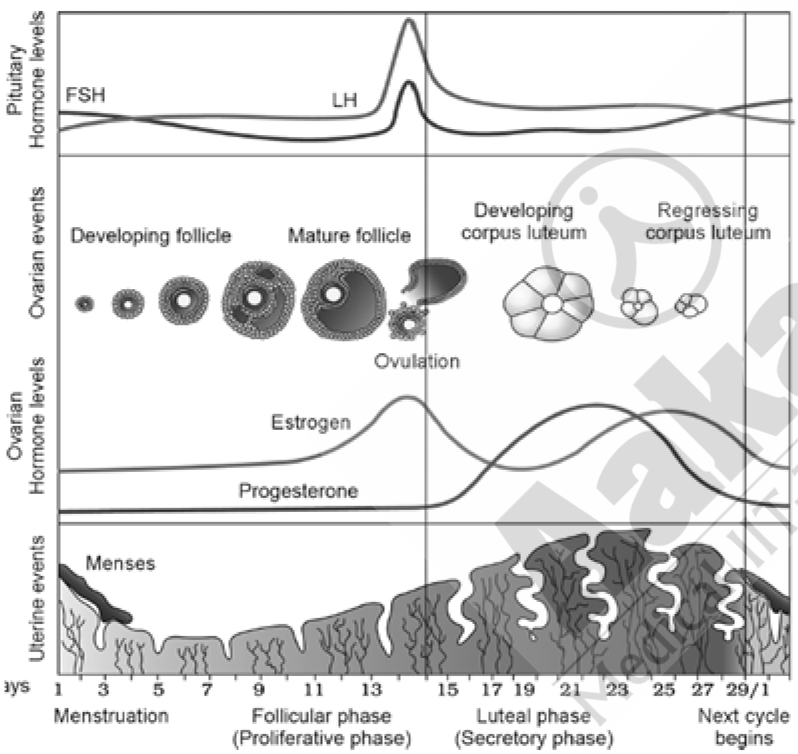
Each restriction endonuclease recognises a specific palindromic nucleotide sequence in the DNA. Some of them cut the strand of DNA a little away from the centre of the palindromic sites.
Restriction enzymes cut each of the two strands of the double helix at specific points in their sugar-phosphate backbone.

(154) Answer : (4)
Solution:
In 1997, the first transgenic cow, Rosie, produced human protein-enriched milk (2.4 grams per litre). The milk contained the human alpha-lactalbumin and was nutritionally a more balanced product for human babies than the natural cow milk.

(155) Answer : (4)
Solution:
hCG stimulates the corpus luteum to secrete progesterone until placenta is formed. Decreased levels of hCG will lead to a drop in the levels of progesterone and spontaneous abortion will occur.

(156) Answer : (4)
Solution:
In some children, ADA deficiency can be cured by bone marrow transplantation; in others, it can be treated by enzyme replacement therapy in which functional ADA is given to the patient by injection. Periodic infusion of genetically engineered lymphocytes can also be used for the management of the disease. However, if the gene isolated from marrow cells producing ADA is introduced into cells at early embryonic stages, it could be a permanent cure.

(157) Answer : (3)
Solution:



(158) Answer : (3)
Solution:
Due to insertional inactivation, *tet*^{R} gene will get inactivated. Thus, recombinants won't survive on tetracycline containing medium. Transformants will grow only on the ampicillin containing medium. Non-transformants will be susceptible to antibiotics as *E.coli* does not carry any antibiotic resistance gene.

(159) Answer : (1)
Solution:
Escherichia is a genus and *coli* is the species. *Eco*RI cuts the DNA between bases G and A only when the sequence 5′-GAATTC-3′ (6 bp) is present in the DNA.

(160) Answer : (4)
Solution:
Elution is a process where separated bands of DNA (after gel electrophoresis) are cut out and extracted from gel piece. Transformation is a procedure through which a piece of DNA is introduced in a host bacterium.

(161) Answer : (2)

Solution:
In polymerase chain reaction, thermostable DNA polymerase (which remains active even at high temperature) is used. It is isolated from the bacterium named, *Thermus aquaticus.*

(162) Answer : (4)
Solution:
In order to force the bacteria to take up the plasmid, the bacterial cell must first be made competent to take up DNA. This is done by treating them with a specific concentration of a divalent cation, such as calcium which increases the efficiency with which DNA enters the bacterium through pores in its cell wall.

(163) Answer : (2)
Solution:
Primary oocyte gets surrounded by a layer of granulosa cells and is called the primary follicle. Tertiary follicle is characterised by a fluid-filled cavity called the antrum. Primary oocyte within the tertiary follicle grows in size and completes its first meiotic division, resulting in the formation of a large haploid secondary oocyte and a tiny first polar body.

(164) Answer : (3)
Solution:
Biolistics - Vector-less method of gene transfer
pBR322 -Artificial plasmid vectors constructed in 1977
β-galactosidase – Reporter enzyme
Agarose gel – Polymeric gel

(165) Answer : (3)
Solution:
Bt-toxin protein exists as inactive protoxin but once an insect ingests the inactive toxin, it is converted into an active form of toxin due to the alkaline pH of the gut which solubilises the protein crystals.

(166) Answer : (2)
Solution:
Cells can be multiplied in a continuous culture system wherein the used medium is drained out from one side while the fresh medium is added from the other to maintain the cells in their physiologically most active log/exponential phase. This type of culturing method produces a larger biomass leading to higher yields of desired protein. The culture is maintained at aerobic condition always.

(167) Answer : (4)
Solution:
Origin of replication (*ori*) is the sequence from where replication starts and is also responsible for controlling the copy number of the linked DNA. Selectable marker is used for identifying and eliminating non-transformants from transformants.

(168) Answer : (1)
Solution:
The sperm head contains an elongated haploid nucleus, the anterior portion of which is covered by a cap-like structure called acrosome, filled with enzymes that help in fertilisation of ovum.

(169) Answer : (2)
Solution:
Since *Hin*d II is not present in any selectable marker gene, then, insertional inactivation of antibiotic resistance encoding gene will not be seen. Hence, recombinants will be amp^R and tet^R.

(170) Answer : (1)
Solution:
Insulin cannot be orally administered to diabetic patients due to its peptide nature, as it can be digested in GI tract because of proteases.

(171) Answer : (3)
Solution:
A single stranded DNA or RNA tagged with a radioactive molecule (probe) is allowed to hybridise to its complementary DNA in a clone of cells followed by its detection using autoradiography.

(172) Answer : (4)
Solution:
Three basic steps in genetically modifying an organism are
(i) Identification of DNA with desirable genes
(ii) Introduction of the identified DNA into the host
(iii) Maintenance of introduced DNA in the host and transfer of the DNA to its progeny

(173) Answer : (1)
Solution:

For normal fertility, 60% of sperms must show normal size and shape and at least 40% of them must show vigorous motility.
60% of 300 million is 180 million
40% of 180 million is 72 million

(174) Answer : (3)
Solution:
Urethra acts as the urinogenital duct in human males as it transports both sperms and urine.
Ureters act as the urinogenital ducts in male frogs.

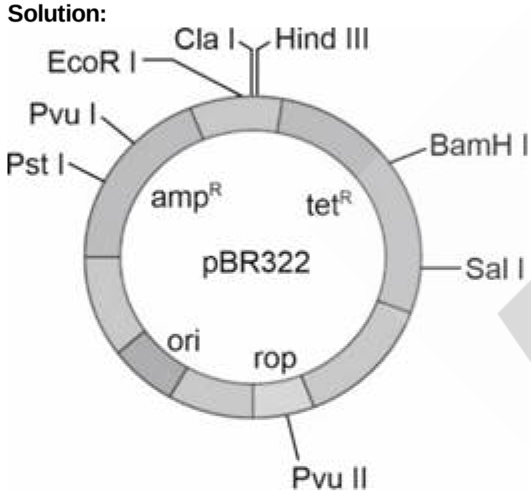
(175) Answer : (4)
Solution:
In gel electrophoresis, since DNA fragments are negatively charged molecules, they can be separated by forcing them to move towards the anode under an electric field through a medium/matrix.

(176) Answer : (3)
Solution:
When cut by the same restriction enzyme, the resultant DNA fragments have the same kind of sticky ends and can be joined by using DNA ligases.

(177) Answer : (1)
Solution:
In human beings, after one month of pregnancy, the embryo's heart is formed. By the end of 24 weeks, the body is covered with fine hair, eye-lids separate and eye-lashes are formed. Cortisol, oxytocin and estrogen mainly participate in the foetal-ejection reflex.

(178) Answer : (2)
Solution:
In the year 1963, the two enzymes responsible for restricting the growth of bacteriophages in *E. coli* were isolated. One of them added methyl group to the bacteria's DNA, while the other cut the DNA of the bacteriophage.

(179) Answer : (1)
Solution:



(180) Answer : (4)
Solution:
Entry of sperm into the cytoplasm of ovum induces the completion of second meiotic division of the haploid secondary oocyte. The second meiotic division is unequal and results in the formation of a haploid polar body and a haploid ovum (ootid).